

Q1(a) $\det(A) = -4$, $\det(B) = 2$

(i) $\det(AB) = \det(A)\det(B) = (-4)(2) = -8$

(ii) $\det(B^T A^T) = \det(B)\det(A) = (2)(-4) = -8$

(b) overlapping line or plane.
coincident

(c) unique solution.

(d) $\begin{bmatrix} 6 & -4 & | & b_1 \\ 3 & -2 & | & b_2 \end{bmatrix} \sim \begin{bmatrix} 6 & -4 & | & b_1 \\ 6 & -4 & | & 2b_2 \end{bmatrix} \xrightarrow{2R_2} \begin{bmatrix} 6 & -4 & | & b_1 \\ 0 & 0 & | & -b_1 + 2b_2 \end{bmatrix} \xrightarrow{R_1+R_2} \begin{bmatrix} 6 & -4 & | & b_1 \\ 0 & 0 & | & -b_1 + 2b_2 \end{bmatrix}$
 $\Rightarrow -b_1 + 2b_2 = 0 \Rightarrow b_2 = b_1/2$

(e) $\det(A^{-1}) = (-2)(-4)(-3)(2) = -48$, $\det(A) = -1/48$

Q1(f)

$[A|I]$

$$\begin{bmatrix} 1 & 2 & -1 & 2 & | & 1 & 0 & 0 & 0 \\ 2 & 4 & 1 & -2 & | & 0 & 1 & 0 & 0 \\ 3 & 6 & 2 & -6 & | & 0 & 0 & 1 & 0 \\ 1 & 3 & 3 & 5 & | & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & 2 & | & 1 & 0 & 0 & 0 \\ 0 & 0 & 3 & -6 & | & -2 & 1 & 0 & 0 \\ 0 & 0 & 5 & -12 & | & -3 & 0 & 1 & 0 \\ 0 & 1 & 4 & 3 & | & -1 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\begin{matrix} -2R_1+R_2 \\ -3R_1+R_3 \\ -R_1+R_4 \end{matrix}}$$

$$\begin{bmatrix} 1 & 2 & -1 & 2 & | & 1 & 0 & 0 & 0 \\ 0 & 1 & 4 & 3 & | & -1 & 0 & 0 & 1 \\ 0 & 0 & 5 & -12 & | & -3 & 0 & 1 & 0 \\ 0 & 0 & 3 & -6 & | & -2 & 1 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_4}$$

$$\begin{bmatrix} 1 & 2 & -1 & 2 & | & 1 & 0 & 0 & 0 \\ 0 & 1 & 4 & 3 & | & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -4/5 & | & -3/5 & 0 & 1/5 & 0 \\ 0 & 0 & 3 & -6 & | & -2 & 1 & 0 & 0 \end{bmatrix} \xrightarrow{\frac{1}{5}R_3}$$

$$\begin{bmatrix} 1 & 2 & -1 & 2 & | & 1 & 0 & 0 & 0 \\ 0 & 1 & 4 & 3 & | & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -4/5 & | & -3/5 & 0 & 1/5 & 0 \\ 0 & 0 & 0 & 4/5 & | & -1/5 & 1 & -3/5 & 0 \end{bmatrix} \xrightarrow{-3R_3+R_4}$$

$$\begin{bmatrix} 1 & 2 & -1 & 2 & | & 1 & 0 & 0 & 0 \\ 0 & 1 & 4 & 3 & | & -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -4/5 & | & -3/5 & 0 & 1/5 & 0 \\ 0 & 0 & 0 & 1 & | & -1/6 & 5/6 & -1/2 & 0 \end{bmatrix} \xrightarrow{\frac{5}{6}R_4}$$

$$\begin{bmatrix} 1 & 2 & -1 & 0 & | & 4/3 & -5/3 & 1 & 0 \\ 0 & 1 & 4 & 0 & | & -1/2 & -5/2 & 3/2 & 1 \\ 0 & 0 & 1 & 0 & | & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 & | & -1/6 & 5/6 & -1/2 & 0 \end{bmatrix} \xrightarrow{\begin{matrix} -2R_4+R_1 \\ \frac{2}{5}R_4+R_2 \\ \frac{12}{5}R_4+R_3 \end{matrix}}$$

$$\begin{bmatrix} 1 & 2 & 0 & 0 & | & 1/3 & 4/3 & 0 & 0 \\ 0 & 1 & 0 & 0 & | & 7/2 & -11/2 & 11/2 & 1 \\ 0 & 0 & 1 & 0 & | & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 & | & -1/6 & 5/6 & -1/2 & 0 \end{bmatrix} \xrightarrow{\begin{matrix} R_3+R_1 \\ -4R_3+R_2 \end{matrix}}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & | & 20/3 & 64/3 & -11 & -2 \\ 0 & 1 & 0 & 0 & | & 7/2 & -11/2 & 11/2 & 1 \\ 0 & 0 & 1 & 0 & | & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 & | & -1/6 & 5/6 & -1/2 & 0 \end{bmatrix} \xrightarrow{-2R_2+R_1}$$

$$A^{-1} = \begin{bmatrix} 20/3 & 64/3 & -11 & -2 \\ 7/2 & -11/2 & 11/2 & 1 \\ -1 & 2 & -1 & 0 \\ -1/6 & 5/6 & -1/2 & 0 \end{bmatrix}$$

Q2(a)

$$T(e_1) = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, T(e_2) = \begin{bmatrix} -1 \\ 6 \end{bmatrix}, A = \begin{bmatrix} 2 & -1 \\ 5 & 6 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 5 \\ -3 \end{bmatrix}\right) = \begin{bmatrix} 2 & -1 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} 5 \\ -3 \end{bmatrix} = \begin{bmatrix} 13 \\ 7 \end{bmatrix}$$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2 & -1 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 - x_2 \\ 5x_1 + 6x_2 \end{bmatrix}$$

2(b) $T(x, y) = (2x + 4xy, y - x)$
Not linear because of $4xy$.

For example,

$$u = (1, 2), c = 2$$

$$T(cu) = T(2, 4) = (4 + 32, 2) = (36, 2)$$

$$c(T(u)) = 2T(1, 2) = 2(10, 1) = (20, 2)$$

$$\Rightarrow c \cdot H \cdot S \neq R \cdot H \cdot S$$

Q3(a)

Nodes:

A

B

Y

Z

Flow in

= Flow out

$$30$$

$$= x_1 + x_2$$

$$x_1 + x_3$$

$$= x_4 + 15$$

$$x_2 + 20$$

$$= x_3 + x_5$$

$$x_4 + x_5$$

$$= 35$$

$$x_1 + x_3 + x_5 = 50$$

$$x_2 - x_3 - x_5 = -20$$

$$x_4 + x_5 = 35$$

general solution:

$$x_1 = 50 - s - t$$

$$x_2 = -20 + s + t$$

$$x_3 = s$$

$$x_4 = 35 - t$$

$$x_5 = t$$

$$\Rightarrow x_1 + x_2 = 30$$

$$x_1 + x_3 - x_4 = 15$$

$$-x_2 + x_3 + x_5 = -20$$

$$x_4 + x_5 = 35$$

Using Gauss Jordan

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 30 \\ 1 & 0 & 1 & -1 & 0 & 15 \\ 0 & -1 & 1 & 0 & 1 & 20 \\ 0 & 0 & 0 & 1 & 1 & 35 \end{array} \right]$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 30 \\ 0 & -1 & 1 & -1 & 0 & -15 \\ 0 & -1 & 1 & 0 & 1 & 20 \\ 0 & 0 & 0 & 1 & 1 & 35 \end{array} \right] \xrightarrow{-R_1 + R_2}$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 30 \\ 0 & 1 & -1 & 1 & 0 & 15 \\ 0 & -1 & 1 & 0 & 1 & 20 \\ 0 & 0 & 0 & 1 & 1 & 35 \end{array} \right] \xrightarrow{-R_2}$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 30 \\ 0 & 1 & -1 & 1 & 0 & 15 \\ 0 & 0 & 0 & 1 & 1 & 35 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{-R_3 + R_4}$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 30 \\ 0 & 1 & -1 & 0 & -1 & -20 \\ 0 & 0 & 0 & 1 & 1 & 35 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{-R_3 + R_2}$$

$$\left[\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 1 & 50 \\ 0 & 1 & -1 & 0 & -1 & -20 \\ 0 & 0 & 0 & 1 & 1 & 35 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{-R_2 + R_1}$$

$$\textcircled{b} x_2 = 15, x_3 = 0$$

$$15 = -20 + 0 + x_5$$

$$x_5 = 35$$

$$\Rightarrow x_4 = 0$$

$$x_1 = 50 - 0 - 35$$

$$x_1 = 15$$

The traffic flow is

$$x_1 = 15$$

$$x_2 = 15$$

$$x_3 = 0$$

$$x_4 = 0$$

$$x_5 = 35$$